Rio Puerco at Cuba — USGS Data Research Brief

Dataset: USGS station 08332525 | Site type: stream | Location: Cuba, Sandoval County, NM

Bottom line: Directly relevant to Cuba, but not its municipal groundwater supply

Unlike the earlier countywide well datasets, this monitoring station is located at Cuba on the Rio Puerco. It directly documents local surface-water conditions. It does NOT measure municipal groundwater levels, aquifer depletion, or drinking-water withdrawals; the Rio Puerco should not be assumed to supply Cuba's municipal wells.

What the three CSV files contain

- monitoring-location-metadata.csv — one station record with coordinates (36.01056, −106.98056), stream classification and Rio Puerco watershed information.
- field-measurements.csv — four discrete gage-height observations (2019–2024), in feet; gage height is river stage relative to a local reference, not groundwater depth.
- channel-measurements.csv — ten discrete channel-flow observations (2019–2026), in ft^3/s; six are zero, and four are nonzero (6.89–20.40 ft^3/s). These are spot measurements, not a continuous streamflow record.

Research-question fit

Question	Support	Use / limitation
Q7 — Groundwater–surface water relationships	Partial	Direct local stream observations; cannot establish groundwater–stream exchange without nearby groundwater data.
Q9 — Water-resource stress dashboard	Moderate	Useful local surface-water indicator; flag sparse, irregular sampling and zero-flow observations.
Q8 — Monitoring coverage and gaps	Moderate	Maps a Cuba-area stream station and demonstrates missing continuous/daily records.
Q6 — Acequias and surface-water sources	Limited	Documents Rio Puerco flow at Cuba, but does not identify acequias or establish municipal supply.
Q1–Q4 — Groundwater trends, wells, depletion, geology	Not supported	Stream measurements cannot substitute for groundwater-level, well-construction, or aquifer data.

Recommended plots for the report

1. Scatter plot of measured channel flow (ft^3/s) versus date, with individual observations only (do not interpolate a continuous hydrograph). 2. Small table/bar chart showing six zero-flow versus four nonzero-flow visits. 3. Location map showing the Rio Puerco station relative to Cuba and any available municipal wells.

Interpretation cautions and next step

Zero-flow readings show no measured flow at the time of those visits—not year-round dryness or a long-term trend. Four stage observations are too few for a reliable stage trend. Seek additional Rio Puerco measurements or nearby stations and pair them with Cuba-specific municipal well/pumping records before drawing conclusions about water-supply sustainability.

Source: map showing measurement locations https://www.waterqualitydata.us/provider/NWIS/USGS-NM/USGS-08332525/

Source: USGS Water Data for the Nation, "Rio Puerco at Cuba, NM," station 08332525; three accompanying CSV exports.